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A Hybrid Deep Learning and Parametric Feature Framework for Similarity Detection in Industrial Technical Drawings

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Abstract: The rapid digitalization of manufacturing has resulted in the accumulation of extensive archives of industrial technical drawings, particularly in mold manufacturing and aluminum profile design, where the efficient retrieval of geometrically and functionally similar designs has become essential for reducing engineering lead time, minimizing design redundancy, and improving product development efficiency. Conventional document management systems rely primarily on metadata-based indexing and keyword matching, making the geometric content and engineering attributes embedded in technical drawings largely inaccessible during retrieval, leading to unnecessary redesign efforts and increased development costs. To address these limitations, a hybrid similarity detection framework was proposed to integrate deep-learning-based visual feature extraction with parametric engineering information. Visual characteristics were extracted using a pre-trained Residual Network (ResNet) model, specifically ResNet-18, a convolutional neural network (CNN), from which 512-dimensional feature vectors were generated to represent the geometric characteristics of technical drawings. Simultaneously, optical character recognition (OCR) combined with rule-based information extraction was employed to identify critical engineering parameters, including profile number, cross-sectional area, and unit weight. The complementary visual and parametric representations were subsequently fused through a hybrid scoring mechanism to improve retrieval robustness. The proposed framework extends conventional content-based image retrieval by integrating domain-specific engineering knowledge. Experimental evaluation conducted on a real industrial dataset containing 2,390 aluminum profile technical drawings demonstrated high retrieval accuracy while maintaining computational efficiency, achieving a vector database search time of approximately 0.1 s and an end-to-end retrieval time (including feature extraction, parametric analysis, and hybrid ranking) of approximately 0.9 s per query. The results indicate that the proposed framework provides an effective solution for large-scale technical drawing management by facilitating design reuse, reducing repetitive engineering work, accelerating product development cycles, and improving operational efficiency in industrial design repositories.

Keywords: Technical drawing analysis; Convolutional neural network; Content-based image retrieval; Similarity detection; Optical character recognition

1 Introduction

In today's highly competitive manufacturing environment, digital transformation has fundamentally changed how engineering knowledge is created, stored, and leveraged. Companies in mold production, machinery manufacturing, and aluminum profile extrusion now maintain massive digital archives containing thousands or even tens of thousands of technical drawings. These documents represent not only geometric designs but also the accumulated intellectual capital of the organization built over many years. Consequently, the ability to rapidly locate and reuse existing designs has become a critical competitive advantage. Although digital transformation continues across the industry, two-dimensional technical drawings remain the dominant format, leading to large-scale archives [1]. In sectors such as mold manufacturing, machine parts, and profile design, quickly finding previously created parts is essential for reducing design time and costs.

When a new customer order or design revision request arrives, engineers usually search the company archive for similar components. In many cases, a geometrically similar or nearly identical part already exists. However, the lack of intelligent search tools often forces engineers to create new designs from scratch. This repetition increases project costs, extends delivery times, creates inconsistencies in production, and unnecessarily expands the number of unique parts in inventory. Traditional document management systems commonly used in industry fall short in addressing these challenges. Most rely solely on metadata such as file name, part number, project code, or creation date, and cannot analyze the visual content, geometric features, or engineering parameters within the drawings [2]. As a result, even when similar parts exist in the archive, redundant redesign remains a frequent problem [3]. Manual inspection of thousands of drawings is impractical, creating a major bottleneck in engineering workflows. This issue is particularly costly in high-variety sectors like aluminum profile manufacturing, where efficient design reuse can provide significant advantages in both cost and speed.

To address these limitations, this study proposes a hybrid similarity detection system that combines deep-learning-based visual analysis with parametric engineering data mining. The system treats each drawing as a multi-layered data source containing both geometric visual information and critical engineering parameters. High-level feature vectors are extracted using Residual Network (ResNet) model, specifically ResNet-18, in the visual branch, while key parameters such as profile number, cross-sectional area, unit weight, and main dimensions are obtained through optical character recognition (OCR) and text processing techniques in the parametric branch. These complementary data sources are integrated via an intelligent hybrid scoring mechanism to deliver more accurate and practical results. The primary objective of this research is to reduce the time engineers spend searching for similar designs in large archives and to maximize the reuse of existing corporate knowledge. By enabling fast and reliable retrieval, the system aims to shorten design cycles, lower engineering and production costs, reduce design errors, and improve overall operational efficiency. Ultimately, it seeks to transform passive technical drawing archives into active, intelligent decision-support tools.

The remainder of this study is structured as follows: Section 2 provides a literature review on technical drawing analysis, content-based image retrieval, and convolutional neural network (CNN)-based feature extraction. Section 3 presents the proposed hybrid methodology in detail, including preprocessing, visual feature extraction, parametric data extraction, similarity metrics, and the hybrid scoring algorithm. Section 4 describes the experimental setup, dataset, and results. Finally, Section 5 discusses the findings, limitations, and future research directions.

2 Literature and Theory

Technical drawings are fundamental communication tools in engineering design, conveying design intent, geometric specifications, tolerances, and production requirements. In fields such as mechanical manufacturing, architecture, and industrial design, these documents contain critical information about product realization. Today, many organizations maintain extensive archives of technical drawings, making rapid analysis and retrieval increasingly important. However, most drawings are stored in portable document format (PDF) or raster image formats, which complicates automatic information extraction [4]. Traditional analysis methods relied primarily on rule-based algorithms and hand-crafted features. While effective for detecting specific symbols or simple geometries, these approaches struggle with the complexity and variability of real-world technical drawings, where text, symbols, and line-based geometries are often intertwined [5]. In recent years, deep learning techniques have significantly advanced the automatic processing of such documents. CNNs have proven particularly successful in recognizing shapes, lines, and symbols by automatically learning hierarchical features from images [6].

These CNN-based models enable effective similarity analysis between different parts and drawing types, facilitating the retrieval of similar designs from large databases [7]. Content-based image retrieval systems further support this process by allowing engineers to search using a reference image. Such systems convert drawings into feature vectors and rank results according to similarity metrics [8, 9]. Beyond visual content, textual information in technical drawings plays a vital role in engineering analysis. Parameters such as profile numbers, cross-sectional area, unit weight, and key dimensions are essential for production decisions. Consequently, modern systems increasingly adopt hybrid approaches that integrate computer vision techniques for feature extraction with text processing methods for parameter identification [4].

Several studies have focused on automatic information extraction and structured data generation from technical drawings, including the detection of views, dimensions, and annotations [5, 7]. Deep-learning-based feature extraction has enabled the conversion of drawings into numerical vectors, allowing efficient similarity computation [6]. Overall, technical drawing analysis has evolved into a multidisciplinary field combining computer vision, machine learning, and natural language processing. These advances offer substantial benefits to manufacturing companies with large archives and are expected to support more sophisticated applications such as error detection, design recommendation, and intelligent retrieval in the near future [4–8].

2.1 Content-Based Image Retrieval

Content-based image retrieval systems automatically locate visually similar images within large collections by analyzing their visual features rather than relying on file names or textual descriptions. Each image is converted into a numerical feature vector through the extraction of attributes such as color, texture, shape, edges, and higher-level semantic characteristics, and the retrieval process relies on comparing these vectors for similarity. Content-based image retrieval pipelines generally follow three stages: feature extraction, feature representation and indexing, and similarity measurement and ranking. First, distinctive characteristics are extracted from the image; second, these characteristics are stored in an indexed database; and finally, the distance between the query vector and the stored vectors is computed to present the closest matches to the user.

In their survey of content-based image retrieval systems, Liu et al. [9] demonstrate that while early implementations relied primarily on low-level descriptors, these methods failed to capture the semantic meaning of images, giving rise to what is known as the 'semantic gap'—a mismatch between extracted visual features and human perception that constrains retrieval accuracy. The emergence of deep learning brought a substantial shift to this field. Maji and Bose [10] demonstrated that CNNs can automatically derive more abstract representations from raw images. In contrast to manually engineered descriptors, these learned representations capture intricate data patterns and yield more discriminative vectors, leading to improved similarity computation, particularly for large datasets.

Wan et al. [11] offered an extensive examination of deep-learning-based retrieval approaches, covering feature extraction strategies, transfer learning, fine-tuning techniques, and how different distance metrics influence outcomes. Their findings indicate that intermediate-layer activations from networks pre-trained on large datasets such as ImageNet yield robust representations for retrieval tasks and that both normalization choices and metric selection have a direct impact on overall performance. Similarity in content-based image retrieval is typically assessed through Euclidean distance (L2), Manhattan distance (L1), or cosine similarity, with cosine similarity frequently cited as offering greater stability for deep-feature vectors in high-dimensional spaces—though the optimal choice can vary by application. Within industrial technical drawing archives, retrieval tasks differ notably from those involving conventional image collections: drawings are largely grayscale or black-and-white with minimal color variation and consist mainly of line-based content, so color cues lose relevance while shape and contour information become essential. Deep architectures can capture the underlying geometric patterns of such drawings, producing representations well suited to identifying similar profiles or cross-sectional geometries.

Overall, the reviewed literature reflects a clear progression from hand-crafted descriptors toward automatic, deep-learning-driven feature extraction. Liu et al. [9] provided a comprehensive survey of underlying content-based image retrieval, Maji and Bose [10] highlighted the advantages of learned features over traditional ones, and Wan et al. [11] provided a systematic account of architectural and performance considerations in modern systems. Building on these insights, the hybrid approach proposed in this study integrates deep-learning-based visual feature extraction with parametric engineering data specific to technical drawings, aiming for a more engineering-oriented and accurate similarity analysis than conventional content-based image retrieval approaches allow.

2.2 Convolutional Neural Network-Based Feature Extraction

Extracting meaningful representations from image data remains a central challenge in computer vision. Traditional approaches relied on manually designed algorithms to identify edges, corners, or textures, whereas recent work has increasingly turned to CNN-based methods [12]. Through their multi-layer architecture, CNNs capture spatial relationships at varying levels of abstraction: shallower layers tend to detect simple patterns such as edges, while deeper layers encode more complex, high-level structures such as object parts [13]. These hierarchical representations support tasks, including classification, recognition, and similarity-based retrieval. A common practice in CNN-based feature extraction involves leveraging pre-trained networks, whose representations learned on large datasets often transfer effectively to new domains—an approach referred to as transfer learning, which has proven beneficial across numerous applications [14]. Models trained on ImageNet, for instance, are frequently repurposed as general-purpose feature extractors for other image analysis tasks.

Among architectures used for image representation, ResNet stands out for its residual connections, which mitigate the vanishing-gradient issue common to conventional deep networks and thereby enable substantially deeper architectures to be trained effectively [15]. Owing to this advantage, ResNet-based models are widely adopted as feature extractors across diverse visual analysis tasks. In retrieval applications, the resulting feature vectors are compared using cosine similarity or other distance metrics to assess image similarity [16], enabling fast and effective search across large image databases. For technical drawings specifically, CNN-based extraction offers clear benefits: such drawings predominantly contain geometric shapes, contours, and structural patterns, and CNNs can learn to encode these elements into vectors that reflect structural correspondences between drawings. This capability supports the automatic identification of similar parts or profiles within technical drawing archives.

In the present study, visual data extracted from technical drawings are transformed into numerical feature vectors using a CNN-based model, after which similarity measurement methods are applied to identify comparable drawings

within the database—offering a more automated and adaptable workflow than traditional approaches.

3 Methodology

This study proposes a hybrid data mining architecture that performs similarity analysis on technical drawing archives by jointly leveraging visual and parametric data (Figure 1). Rather than treating technical drawings merely as static documents, the system regards them as multi-layer data sources, analyzing both their visual characteristics and the textual engineering parameters embedded within them. The proposed approach comprises four main stages: data preprocessing, visual feature extraction, parametric data extraction, and similarity evaluation.

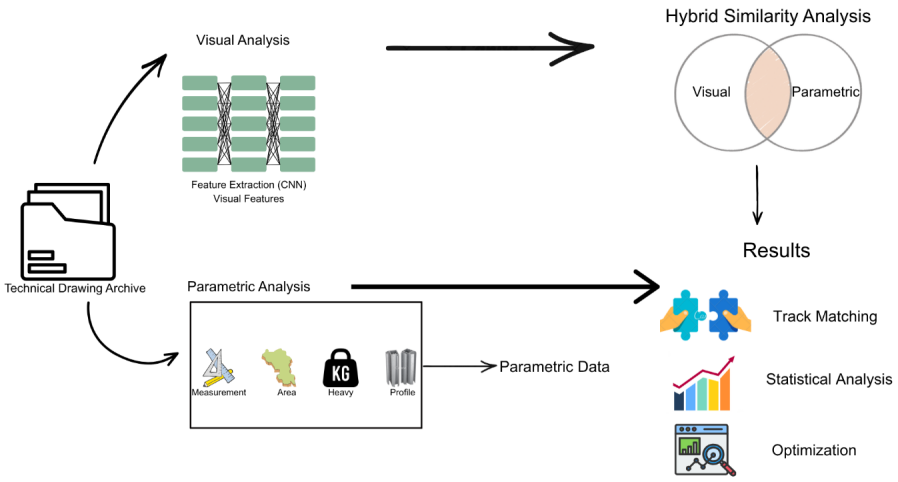


Figure 1. Architecture of the proposed hybrid similarity detection framework

Note: CNN = convolutional neural network.

3.1 Hybrid Data Processing Architecture

The system architecture relies on two parallel processing branches, enabling the extraction of both visual and textual information from technical drawings. As for the visual analysis branch (artificial intelligence-based analysis), images derived from the drawings are processed through deep learning techniques, where a CNN architecture generates high-dimensional feature vectors. These vectors form a numerical representation of the geometric structure and shape characteristics of each drawing. As for the parametric analysis branch (text-based analysis), engineering parameters such as area, weight, profile number, and measurement values are extracted through text processing techniques. OCR combined with regular expressions converts this information into a structured format for inclusion in the similarity analysis. Combining these two data sources yields more reliable outcomes than relying solely on visual similarity.

3.2 Image Preprocessing and Noise Reduction

Technical drawings typically contain additional elements such as title blocks, dimension lines, text annotations, and frame components, which can complicate analysis of the main profile geometry. To address this, image preprocessing is applied prior to visual analysis. PDF drawings are first converted into raster images, followed by noise reduction and thresholding. Otsu's thresholding method [17] sharpens the principal contours, after which contour analysis identifies the core profile geometry while filtering out extraneous graphic elements. This preprocessing step allows the CNN-based model to concentrate on the relevant geometric structures, thereby improving feature extraction accuracy.

3.3 Convolutional Neural Network-Based Visual Feature Extraction

Visual similarity analysis requires high-level features derived from technical drawing images, for which deep-learning-based CNN architectures are employed. CNNs can learn spatial relationships within images and automatically capture visual characteristics such as edges, contours, and geometric structures. A pre-trained CNN model is used for feature extraction here; having been trained on large datasets, such models offer strong representational capacity across different image domains through transfer learning. Activation values from the model's intermediate layers form a high-dimensional feature vector for each drawing, which is then stored in a database for use in similarity calculations.

Visual feature extraction in this work relies on ResNet-18, a deep CNN. Its residual (skip) connection structure mitigates the vanishing gradient problem common in deep networks by allowing input data to bypass certain layers and pass directly to later ones, enabling deeper networks to train more stably. The model is used with pre-trained weights, with its final fully connected classification layer removed so that it functions purely as a feature extractor—producing a vector representation of visual content rather than a classification output. The network's initial convolutional layers detect low-level features such as edges, corners, and lines, while batch normalization layers normalize the data to support learning. The rectified linear unit (ReLU) activation function introduces non-linearity, helping the model capture more complex patterns, and max pooling layers reduce feature map dimensions, lowering computational cost and encouraging more generalized feature learning. The residual blocks at the core of the architecture combine multiple convolutional layers with skip connections that add input directly to output, allowing the network to grow deeper and represent complex geometric structures more effectively. Finally, global average pooling condenses the feature maps into a single 512-dimensional vector per image, providing a compact representation of each drawing's visual content for similarity analysis.

Once feature vectors are obtained, they serve as the basis for comparing the query drawing against those in the database. Cosine similarity is applied for this comparison, given its stability in high-dimensional spaces and widespread use in image retrieval. Experimental results indicate that this CNN-based approach effectively captures geometric structures in technical drawings, with drawings sharing similar profile geometries—particularly contours and overall shapes—receiving high similarity scores. Nonetheless, reliance on visual similarity alone can occasionally lead to mismatches; for instance, drawings of differing sizes but similar shapes may be incorrectly matched due to scale variation.

By leveraging intermediate layer representations, the deep architecture acts as a nonlinear function approximator. The stacked convolutional layers with ReLU activations and residual skip connections transform complex geometric contours into a high-dimensional feature space, capturing intricate shape similarities that traditional linear descriptors cannot model.

3.4 Parametric Data Extraction

Beyond visual geometry, technical drawings also encode important engineering parameters. Accordingly, the second analysis branch extracts textual information through OCR-based detection, identifying numerical values within the drawings. Regular expressions are then applied to isolate engineering parameters such as area, unit weight, profile number, and measurement values, converting them into a structured format that is stored in a database and serves as an additional validation layer within the similarity analysis.

3.5 Similarity Measurement and Vector Space Model

Feature vectors derived from visual extraction are compared within a high-dimensional vector space, where each drawing is represented by its own vector and similarity is determined by the distance or angular relationship between vectors. Four metrics are applied for this purpose: cosine similarity, Euclidean distance (L2), Manhattan distance (L1), and Minkowski distance (Lp). While cosine similarity captures angular relationships between vectors [18], the distance-based metrics quantify geometric separation, with cosine similarity generally regarded as more robust in high-dimensional spaces. Together, these metrics provide complementary perspectives on similarity, and their outputs feed into the system's hybrid scoring mechanism.

3.5.1 Cosine similarity

Cosine similarity quantifies the angular relationship between two feature vectors and is widely adopted in content-based image retrieval, particularly for high-dimensional data. Because it captures direction rather than magnitude, it yields stable results for technical drawing comparisons—values closer to 1 indicate greater similarity between two drawings.

$$\cos(\theta) = \frac{\overrightarrow{A} \cdot \overrightarrow{B}}{\|A\| \, \|B\|} \tag{1}$$

where, θ represents the angle between the two feature vectors; A and B represent the 512-dimensional feature vectors extracted by ResNet-18 from the query drawing and a database drawing, respectively; $A \cdot B$ denotes the dot product of vectors A and B; and $\|A\|$ and $\|B\|$ represent the Euclidean norms (magnitudes) of vectors A and B, respectively.

3.5.2 Euclidean distance

Euclidean distance is among the most widely used measures of geometric separation between feature vectors, computed by summing the squared differences of corresponding vector components and taking the square root of

that sum. Smaller distances, approaching 0, indicate greater similarity between drawings.

$$d(A, B) = \sqrt{\sum_{i=1}^{n}(A_i - B_i)^2} \tag{2}$$

where, $d(A, B)$ denotes the Euclidean distance between feature vectors A and B; A_i and B_i represent the i-th scalar components of vectors A and B, respectively; i is the feature component index ($i = 1, 2, \ldots, n$); and n is the total dimension of the feature vectors ($n = 512$).

3.5.3 Manhattan distance

Manhattan distance is computed as the sum of absolute differences between two vectors and can produce more stable results than Euclidean distance in certain high-dimensional spaces. Its name derives from its resemblance to measuring distances along city blocks.

$$d(A, B) = \sum_{i=1}^{n} |A_i - B_i| \tag{3}$$

where, $d(A, B)$ denotes the Manhattan distance between vectors A and B; A_i and B_i represent the i-th components of vectors A and B; i is the feature index; and n is the vector dimension ($n = 512$).

3.5.4 Minkowski distance

Minkowski distance generalizes both Euclidean and Manhattan distances, with the resulting metric depending on the parameter p—yielding Manhattan distance when $p = 1$ and Euclidean distance when $p = 2$. This metric is also examined in this study to explore alternative similarity measures.

$$d(A, B) = \left(\sum_{i=1}^{n} |A_i - B_i|^p \right)^{1/p} \tag{4}$$

where, $d(A, B)$ denotes the Minkowski distance between vectors A and B; A_i and B_i are the i-th vector components; n is the vector dimension ($n = 512$); and p is the order parameter defining the metric space ($p = 1$ for Manhattan distance and $p = 2$ for Euclidean distance).

3.6 Hybrid Similarity Score

A central component of the proposed system is its hybrid similarity score—a mechanism that goes beyond visual similarity to incorporate engineering parameters extracted from technical drawings. Visual similarity forms the foundation of the score, with additional points awarded when parametric values such as cross-sectional area, unit weight, and profile number align within specified tolerances, yielding more dependable results overall.

The hybrid similarity score is expressed as:

$$\begin{aligned} S_{\text{hybrid}} &= S_{\text{visual}} + B_{\text{area}} + B_{\text{weight}} + B_{\text{profile}} \\ S_{\text{visual}} &= 100 \times \cos(\theta) = 100 \times \frac{A \cdot B}{\|A\| \, \|B\|} \end{aligned} \tag{5}$$

where, S_{hybrid} represents the final hybrid similarity score; S_{visual} is the base visual similarity score derived from the cosine similarity between the 512-dimensional feature vectors A and B, scaled by a factor of 100 ($S_{\text{visual}} \in [0, 100]$); A and B represent the 512-dimensional visual feature vectors extracted by ResNet-18 for the query technical drawing and a database technical drawing, respectively; $\|A\|$ and $\|B\|$ denote the Euclidean norms of vectors A and B; B_{area} is an additional parametric bonus awarded when both cross-sectional area values exceed 1.0 mm^2 and satisfy $|C_{\text{query}} - C_{\text{db}}|/C_{\text{query}} < 0.02$ (2% relative tolerance), adding 50 points, otherwise 0; C_{query} and C_{db} denote the cross-sectional area values of the query and database drawings, respectively; B_{weight} is an additional parametric bonus awarded when both unit weight values exceed 0.1 kg/m and satisfy $|W_{\text{query}} - W_{\text{db}}|/W_{\text{query}} < 0.02$ (2% relative tolerance), adding 50 points, otherwise 0; W_{query} and W_{db} denote the unit weight values of the query and database drawings, respectively; and B_{profile} is an additional bonus of $+200$ points awarded if the string representations of the profile identification numbers match exactly, otherwise 0.

With this hybrid scoring mechanism, visually similar drawings but have different technical properties can be separated more accurately. The tolerance thresholds (2%) and bonus weights (+50, +50, +200) were determined empirically through extensive testing on a real industrial dataset comprising approximately 2,390 aluminum profile technical drawings. These values were manually calibrated to reflect typical manufacturing tolerances while remaining strict enough to exclude irrelevant matches. The substantial bonus assigned to exact profile number matching

reflects the engineering principle that an identical part number generally indicates the same component, possibly with minor revisions. This additive structure allows strong parametric matches to meaningfully boost a result's ranking even when visual similarity is only moderate, while simultaneously preventing visually similar but parametrically distinct drawings from being ranked as close matches.

This scoring formulation introduces conditional, threshold-based logic into the retrieval decision space. Rather than relying on a simple linear combination, the piecewise bonus allocation establishes context-aware decision boundaries, effectively preventing visually similar drawings with distinct engineering parameters from ranking closely.

3.7 Software Infrastructure

The system was developed using the open-source PyTorch and OpenCV libraries. PyTorch [19] is a modern machine learning framework for building and training deep learning models; its dynamic computation graphs and graphics processing unit support enable high-performance image-based feature extraction. OpenCV [20], a comprehensive computer vision and image processing library, was employed for preprocessing steps such as thresholding, contour analysis, and image transformations. Together, these technologies support efficient execution of both the visual feature extraction and similarity analysis processes.

3.8 Implementation Details and Reproducibility

The system was implemented in Python 3.10 using PyTorch 2.0.1, OpenCV 4.8.0, and PyMuPDF. As for image preprocessing, PDF pages were rendered at 120 dots per inch (DPI) and converted to grayscale, then binarized using Otsu's thresholding [17]. The main profile contour was extracted using *RETR_EXTERNAL* and *CHAIN_APPROX_SIMPLE*, and the largest contour was masked to remove title blocks and annotations. Final images were resized to 256 × 256 pixels with center cropping. In terms of visual feature extraction, a pre-trained ResNet-18 model was used with its final classification layer removed; 512-dimensional features were extracted from the global average pooling layer following ImageNet normalization.

In terms of parametric extraction, text was extracted directly from PDF files using PyMuPDF, with regular expressions subsequently applied to capture the profile number, cross-sectional area, and unit weight. As for the hybrid similarity scoring, cosine similarity served as the base score, with additional bonuses of +50 points awarded for area or weight differences under 2%, and +200 points for an exact profile number match.

4 Experimental Results and Analysis

This section evaluates the performance of the proposed hybrid technical drawing similarity system. The analysis examines both the accuracy of similarity detection and the system's behavior when applied to large-scale technical drawing archives. To this end, the system was tested on different datasets, with results assessed using a range of performance metrics.

4.1 Experimental Environment and Dataset

The experiments were conducted on a dataset drawn from the technical drawing archive of an aluminum profile manufacturing company, comprising approximately 2,390 technical drawings that include profile cross-section geometries and production parameters, mostly stored as engineering documents in PDF format. Each drawing contained key engineering information such as profile cross-section geometry, dimensional data, profile number, cross-sectional area, and unit weight. For processing, drawings were first converted into image format and then separated into two distinct representations: visual data for CNN-based feature extraction and parametric engineering data extracted using OCR and text processing methods.

This setup provided a realistic industrial test environment, allowing both geometric similarity analysis and comparison of engineering parameters through a hybrid approach. Accordingly, the system generated two representations for each drawing:

- Visual feature vector: high-dimensional representation extracted by the CNN model
- Parametric data: engineering parameters obtained using OCR and regular expression methods (such as area, weight, etc.)

These two data types were used together to calculate similarity relationships between drawings. The experiments were run on a graphics processing unit-supported computer environment (12th gen Intel Core i7-1255U (12 central processing units), 32 GB random access memory). System performance was evaluated in terms of indexing time, search time, and memory usage.

4.2 System Performance

To evaluate the performance of the proposed system, three main metrics were considered, namely indexing time, search time, and memory usage. Indexing time refers to the duration required during the initial stage, when feature

vectors are generated for each drawing and stored in the database, while search time denotes the interval between submitting a query drawing and retrieving similar results. The experiments demonstrated scalable performance on large datasets: a total of 2,390 drawings were indexed in 441.8 seconds (Figure 2). Because feature vectors were pre-computed and stored, retrieval operations executed very quickly, with average search times observed at the millisecond level. Indexing time was found to increase linearly with dataset size, whereas search time remained relatively constant, indicating that the architecture scales well for large technical drawing archives.

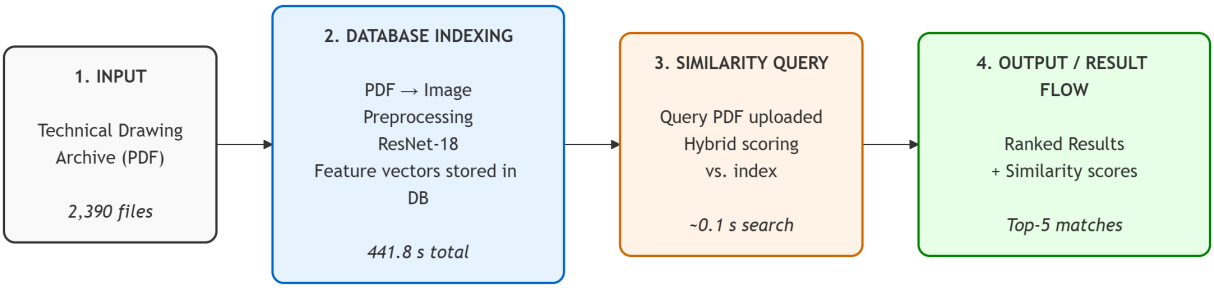


Figure 2. System pipeline across four distinct sequential stages
Note: DB = database; PDF = portable document format; ResNet = Residual Network.

These findings confirm that the developed system can deliver high performance across large-scale archives. Since indexing occurs only once and subsequent searches rely on pre-computed vectors, similarity retrieval times remain minimal. To evaluate system performance, three distinct temporal metrics were formally defined and measured:

Offline Indexing Time: The time required to process the complete dataset of 2,390 drawings and store pre-computed feature vectors in the database, which totaled 441.8 seconds (approximately 0.185 s per drawing).

Query Search Time: The purely computational duration for vector similarity comparison and hybrid score calculation against pre-computed indexed database entries, which executed in approximately 0.1 s (averaging 0.047 ms per profile scan).

End-to-End Processing Time: The total wall-clock duration for a newly submitted technical drawing, encompassing file rendering, preprocessing, ResNet-18 feature extraction, OCR parametric extraction, vector search, and final hybrid result rendering. Across test runs, the system achieved an average end-to-end processing time of approximately 0.9 s per query.

4.3 Visual Similarity Analysis Results

The detailed analysis report generated by the developed interface includes statistical summaries of the metrics used in similarity calculation (Figure 3), reporting the mean, standard deviation, minimum, and maximum values for cosine similarity, Euclidean distance, and Manhattan distance to characterize the distribution of results. The report also includes performance figures such as indexing time, total search time, and per-profile search time. Together, these statistics offer a means of assessing both system performance and the reliability of similarity outcomes, enabling evaluation not just of which drawings are similar but of how statistically meaningful those matches are.

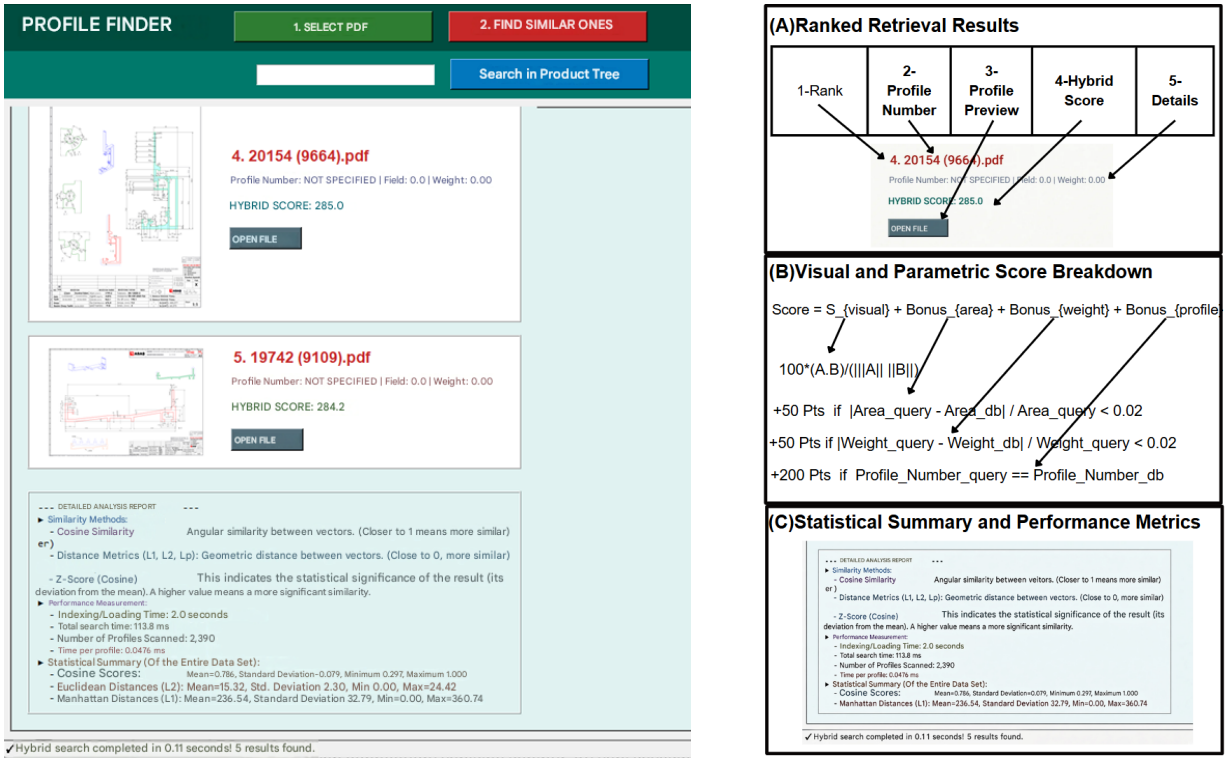
For an example query, cosine similarity values were calculated with a mean of 0.786, a standard deviation of 0.079, a minimum of 0.297, and a maximum of 1.000—indicating that a substantial portion of the dataset exhibits medium-to-high similarity. For Euclidean distance, the mean was 15.32 with a standard deviation of 2.30, while Manhattan distance yielded a mean of 236.54. Considered together, these distance metrics indicate that similarity values between drawings cluster within a defined range, suggesting the system identifies similar profiles in a statistically consistent manner. Overall, these results demonstrate that the visual similarity system produces outcomes that are not only visually meaningful but also numerically and statistically reliable.

4.4 Effect of the Hybrid Scoring Mechanism

The proposed hybrid approach evaluates engineering parameters in addition to visual similarity. Parameters such as cross-sectional area and unit weight play a significant role in this analysis, as they reflect actual production characteristics. The scoring mechanism accounts for both the visual similarity score and the consistency of parametric data: when visual similarity is high and parametric values fall within a defined tolerance, additional points are awarded, raising the result's position in the ranking.

As shown in Table 1 and Table 2, the preliminary experiments compared several CNN models (ResNet-18, ResNet-34, ResNet-50, Contrastive Language–Image Pre-training, DeepSeek) under various parameter settings, including DPI, input size, and center crop, with visual similarity assessed via cosine similarity, Euclidean distance, and Manhattan distance for each configuration. ResNet-50 yielded a relatively low average cosine similarity of 0.696,

whereas preprocessing adjustments such as the CenterCrop-256 transforms and increased DPI raised this value to between 0.784 and 0.785. CenterCrop and higher DPI settings similarly improved per-profile search time and overall query duration. Based on these comparisons, configurations achieving the highest cosine similarity and lowest distance values were selected as reference points for further development. The analysis confirms that both preprocessing choices and model selection have a direct effect on similarity detection performance and on the precision of the hybrid scoring mechanism. It further indicates that the hybrid approach reduces false positive matches, with visually similar drawings that differ in technical properties being more accurately separated—underscoring the advantage of the hybrid method over purely image-based retrieval for engineering applications.



(a) (b)

Figure 3. Outputs generated by the proposed application

Note: L1 = Manhattan distance; L2 = Euclidean distance; Lp = Minkowski distance; PDF = portable document format; pts = points.

Table 1. Preliminary performance evaluation and cosine similarity results of the compared models

Version	Model	Similar Results Count	Initial Indexing Time	Total Search Time	Time per Profile	Average Cosine Similarity	Standard Deviation of Cosine Similarity	Minimum Cosine Similarity	Maximum Cosine Similarity
V1	ResNet-18	3	562	44.9	0.0186	0.770	0.093	0.259	1
V2	Parallel Indexing	3	320.5	51.6	0.0207	0.769	0.092	0.259	1
V3	Low-DPI	3	286.6	45.2	0.0187	0.772	0.091	0.261	1
V4	ResNet-34	2	244.1	44.1	0.0177	0.769	0.083	0.316	1
V5	ResNet-50	3	302.2	75.6	0.0303	0.696	0.147	0.142	1
V6	CenterCrop-256	4	234.5	42.6	0.0171	0.784	0.080	0.297	1
V7	Resize-350	2	231.5	58.9	0.0236	0.772	0.088	0.298	1
V8	High-DPI	4	258.2	44.1	0.0177	0.785	0.083	0.296	1
V9	Contrastive Language–Image Pre-training	2	869.9	46.1	0.0191	0.750	0.103	0.384	1
V10	DeepSeek	3	15,696.0	57.4	0.0237	0.672	0.132	0.253	1

Note: ResNet = Residual Network; DPI = dots per inch.

Among the backbone models evaluated, ResNet-18 was selected as the primary feature extractor for the final system. While deeper architectures such as ResNet-50 and alternatives like Contrastive Language–Image Pre-training performed competitively under certain configurations, ResNet-18 offered the best balance between accuracy,

computational efficiency, and practical constraints for industrial deployment. As shown in Table 1, ResNet-18 configurations (V1, V2, V3) achieved strong cosine similarity scores (averaging around 0.77) with markedly lower indexing times (320–562 seconds) compared to more complex alternatives. ResNet-50 (V5), despite offering richer representations in some respects, incurred higher computational cost—an indexing time of approximately 302 seconds and a per-profile search time of 0.0303 seconds. Contrastive Language–Image Pre-training (V9) and DeepSeek (V10) produced lower average cosine similarity values (0.750 and 0.672, respectively) alongside substantially longer indexing times (869 and 15,696 seconds), making them less practical for large-scale on-premise industrial deployments where fast indexing and limited resource use are essential.

Table 2. Euclidean and Manhattan distance statistics of the compared models

Version	Model	Average Euclidean Distance (L2)	Standard Deviation of Euclidean Distance (L2)	Minimum Euclidean Distance (L2)	Maximum Euclidean Distance (L2)	Average Manhattan Distance (L1)	Standard Deviation of Manhattan Distance (L1)	Minimum Manhattan Distance (L1)	Maximum Manhattan Distance (L1)
V1	ResNet-18	15.23	2.54	0	25.56	236.20	36.79	0	391.75
V2	Parallel Indexing	15.27	2.52	0	25.56	236.94	36.67	0	391.75
V3	Low-DPI	15.25	2.52	0	25.52	236.66	36.66	0	391.84
V4	ResNet-34	18.63	2.81	0	27.85	289.42	39.72	0	436.31
V5	ResNet-50	10.87	2.19	0	18.49	144.55	35.73	0	291.01
V6	CenterCrop-256	15.39	2.31	0	24.42	237.70	33.15	0	360.74
V7	Resize-350	14.24	2.11	0	23.63	229.88	31.16	0	378.83
V8	High-DPI	15.49	2.38	0	24.66	240.18	33.83	0	367.07
V9	Contrastive Language–Image Pre-training	0.69	0.14	0	1.11	12.02	2.16	0	18.24
V10	DeepSeek	0.80	0.15	0	1.22	21.17	3.75	0	31.48

Note: ResNet = Residual Network; DPI = dots per inch.

The residual connections in ResNet-18 enable effective feature extraction even within a relatively shallow architecture, lowering the risk of overfitting on datasets characterized by limited color variation and standardized line-based geometries. This choice supports real-time search performance (approximately 0.9 s) while preserving high accuracy, making it well suited to manufacturing environments operating under constrained hardware resources.

4.5 Statistical Analysis and Z-Score Evaluation

To examine the distribution of search results, a statistical evaluation of similarity scores was conducted, calculating their mean and standard deviation and assessing significance via z-score analysis. The z-score method indicates how significant a given similarity score is relative to the overall dataset distribution, with results deviating further from the mean considered stronger matches. The analysis shows that correct matches tend to exhibit z-scores above the average, suggesting that the system's top-ranked results are statistically meaningful. This provides an additional mechanism for assessing the reliability of search outcomes and the quality of matches.

5 Discussion and Conclusions

This study presents a hybrid data mining approach for the fast and accurate detection of similar designs within industrial technical drawing archives, combining CNN-based visual feature extraction with OCR-supported parametric data analysis. By incorporating engineering parameters such as cross-sectional area and unit weight into an intelligent scoring mechanism with defined tolerance ranges, the approach mitigates the matching errors that purely visual systems often produce due to scale and size discrepancies. Through this integration, technical drawings shift from being passive visual documents to functioning as multi-dimensional decision-support tools. The experimental evaluation yielded strong results overall. Offline indexing of the dataset containing 2,390 technical drawings was completed in 441.8 seconds. During online execution, pure vector-based search across the pre-indexed database required approximately 0.1 s, whereas the complete end-to-end processing pipeline—encompassing raw document ingestion, visual preprocessing, ResNet-18 feature extraction, OCR-based parametric parsing, and final hybrid ranking—was completed in an average of approximately 0.9 s per query. Furthermore, the dataset's average cosine similarity of 0.78 indicates that geometrically related designs are clustered in a statistically consistent manner. From an industrial standpoint, this approach substantially enhances the reuse of corporate knowledge: engineers can retrieve previously approved designs within seconds, eliminating unnecessary redesign efforts, shortening development time, and directly reducing production costs. In effect, archived engineering knowledge is converted into an actively usable information network.

Despite these encouraging outcomes, certain limitations remain. The current system has been evaluated only on two-dimensional technical drawings, and OCR accuracy remains sensitive to scan quality and font type. Future work will focus on extending the framework to support three-dimensional computer-aided design models, integrating natural language processing techniques to incorporate production notes and tolerance information, and exploring cloud-based, scalable architectures suitable for large-scale industrial deployment. Overall, by combining visual deep learning with text-based parametric analysis, this work offers a more engineering-oriented and reliable alternative to traditional content-based retrieval systems, demonstrating the considerable potential of artificial intelligence and data mining techniques in industrial document management and laying a solid foundation for future intelligent engineering assistants.

Author Contributions

Conceptualization, S.K. and A.S.; methodology, S.K.; software, S.K.; validation, S.K. and A.S.; formal analysis, S.K.; investigation, S.K.; resources, S.K.; data curation, S.K.; writing—original draft preparation, S.K.; writing—review and editing, S.K. and A.S.; visualization, S.K.; supervision, A.S. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request. The dataset consists of industrial technical drawings and contains proprietary information owned by the collaborating company; therefore, it is not publicly available.

Conflicts of Interest

The authors declare no conflicts of interest.

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